Score Contributor 2: Granularity and Pacing Score (GPS)

BASE PROMPT:

This is how I had structured the prompt while running with Gemma 4. Please use it for Qwen 235 B

```
"""
granularity_pacing/common.py

Shared prompt, rubric, and output-parsing logic for the "Gran
ularity-Pacing
Metric" — the judge that scores whether the VOLUME and COMPLE
XITY of
elements targeted at a given animation timestep together form
a digestible
unit for a viewer, or overload them.
"""

import os
import re

# -------------------------------------------------------------
----------------
# Band definitions and Style Adapters
# -------------------------------------------------------------
----------------

BAND_LABELS = {
    4: "Well-Paced",
    3: "Minor Overload",
    2: "Moderate Overload",
```

```
    1: "Major Overload",
    0: "Severe Overload",
}

STYLE_ADAPTERS = {
    "Progressive Reveal": (
        "*** STYLE ADAPTER: PROGRESSIVE REVEAL ***\n"
        "In Progressive Reveal, elements are cumulatively drawn onto the canvas.\n"
        "- 'Newly Targeted Elements': For this evaluation, identify the specific diagram elements "
        "that were just drawn or rendered in the CURRENT frame that were not present in the PREVIOUS frame."
    ),
    "Alpha Masking": (
        "*** STYLE ADAPTER: ALPHA MASKING ***\n"
        "In Alpha Masking, the entire diagram is visible, but active elements are fully opaque "
        "while inactive elements are masked (transparent or dimmed).\n"
        "- 'Newly Targeted Elements': For this evaluation, identify the specific diagram elements "
        "that transitioned to FULL OPACITY in the CURRENT frame. (Do not evaluate elements that faded "
        "back to a dimmed state; focus only on what is actively highlighted now)."
    ),
    "Colour Pop": (
        "*** STYLE ADAPTER: COLOUR POP ***\n"
        "In Colour Pop, the entire diagram starts in greyscale, and active elements permanently "
        "transition to full color.\n"
        "- 'Newly Targeted Elements': For this evaluation, identify the specific diagram elements "
        "that transitioned from greyscale to FULL COLOR in the CURRENT frame."
```

```
    ),
    "Hopping Bounding Box": (
        "*** STYLE ADAPTER: HOPPING BOUNDING BOX ***\n"
        "In Hopping Bounding Box, the diagram itself does not
move or change. Instead, a discrete "
        "highlight box jumps to enclose the active region.\n"
        "- 'Newly Targeted Elements': For this evaluation, id
entify the specific underlying diagram "
        "elements (nodes, blocks, etc. from the XML) that are
ENCLOSED by the new position of the "
        "bounding box in the CURRENT frame. You are scoring t
he volume/complexity of the elements inside "
        "the box, not the box itself."
    ),
    "Sliding Bounding Box": (
        "*** STYLE ADAPTER: SLIDING BOUNDING BOX ***\n"
        "In Sliding Bounding Box, the diagram itself does not
move or change. Instead, a highlight "
        "box smoothly slides to enclose the active region.\n"
        "- 'Newly Targeted Elements': For this evaluation, id
entify the specific underlying diagram "
        "elements (nodes, blocks, etc. from the XML) that are
ENCLOSED by the bounding box once it "
        "has come to rest in the CURRENT frame. You are scori
ng the volume/complexity of the elements inside "
        "the box, not the box itself."
    )
}

RUBRIC_TEXT = """
You will score ONE combined severity band (0-4) per timestep,
but you must
reason about the two underlying sub-criteria SEPARATELY befor
e collapsing
them into a single band. Do not skip straight to a number.
```

```
--- SUB-CRITERION 1: VOLUME ---
Judge whether the NUMBER of elements/groups newly targeted/re
vealed together
at THIS timestep is a digestible chunk for a viewer, or an ov
erload.
- OVERLOAD (too many elements dumped/targeted in one step) is
a real violation
  and must be penalized.
- SPARSITY IS NEVER A VIOLATION. Revealing very few elements
per step, or
  even just one, is NOT a crime and must NOT be penalized — b
reaking a
  complex diagram into many small, digestible chunks is exact
ly what good
  pacing looks like. Do not dock points for "too few" element
s, ever.
- EXCEPTION — hopping/sliding single-element focus: if the an
imation uses a
  moving highlight mechanism (e.g. a bounding box, spotlight,
or cursor)
  that visits and focuses on ONE element at a time — even if
many elements
  exist nearby or in that region of the diagram — this is NOT
a volume
  violation. Score Volume band 4 for that step as long as the
actual focus
  of the step is a single element.

--- SUB-CRITERION 2: COMPLEXITY ---
Judge whether the elements newly targeted together at THIS ti
mestep are
individually simple enough, or intrinsically complex enough,
that grouping
them causes overload EVEN IF the raw count (Volume) is accept
able.
Use the XML schema to gauge intrinsic complexity of what's be
```

```
ing targeted:
- A `block` containing several `child_node`s is a heavier sin
gle unit than
  one `standalone_node` — targeting a fully-populated block a
t once can be
  an overload even though it is nominally "one element."
- An element with many incident `edge`s (high fan-in/fan-out)
asks the
  viewer to also absorb several new relationships at once, no
t just the
  element itself — this adds complexity beyond the node in is
olation.
- Two or more elements that are each independently complex (p
er the above)
  being targeted in the SAME timestep is a stronger violation
than one
  complex element plus several simple ones.

--- COMBINING VOLUME AND COMPLEXITY INTO ONE BAND ---
IMPORTANT PRIORITY RULE: Complexity violations are weighted M
ORE heavily
than Volume violations. If you are ever torn between how much
to penalize
a step for its volume versus its complexity, resolve the ambi
guity in favor
of penalizing complexity more — i.e. it is better animation d
esign to break
a step into more, smaller pieces (accepting slightly higher p
er-step
"volume" if truly unavoidable) than to accept higher per-step
complexity.
When scoring, this means: a step with borderline-high complex
ity should
generally be scored at least as harshly as an equivalent bord
erline-high
volume step, and ties should be resolved toward the harsher,
```

```
complexity-
driven interpretation.

4 - Well-Paced: The volume of elements targeted is a digestib
le chunk (or
    intentionally sparse — see above), AND nothing targeted t
ogether is
    individually complex enough to cause overload on its own
or in
    combination with the rest of the group.
3 - Minor Overload: Slightly more elements, or slightly more
complexity,
    than ideal for one step — but a viewer could still follow
it without
    losing track. (Never use this band merely because FEW ele
ments were
    shown — that is not a violation.)
2 - Moderate Overload: The step asks the viewer to absorb a n
oticeably
    large batch of elements, OR mixes in one clearly complex
sub-element
    (per the XML complexity signals above) alongside others,
such that some
    elements would likely be glossed over — but the step is n
ot fully
    overwhelming.
1 - Major Overload: EITHER a large volume of elements is dump
ed into one
    step with no digestible boundary, OR two or more genuinel
y complex
    elements (each independently warranting its own explanato
ry step) are
    clubbed together, causing real viewer overload.
0 - Severe Overload: The step targets an unreasonably large n
umber of
    elements and/or bundles multiple high-complexity sub-stru
```

```
ctures
    together such that the highlight is effectively "the whol
e diagram (or a
    large complex chunk of it) at once" — there is no meaning
ful pacing at
    this step.

Tie-break rule:
1. If both Volume and Complexity are band 4, the combined ban
d is 4.
2. If only ONE of the two sub-criteria is below 4, the combin
ed band is
   normally that sub-criterion's band.
3. If BOTH sub-criteria are below 4 (volume and complexity bo
th contribute
   to overload at this step), anchor the combined band on the
COMPLEXITY
   band, then go ONE band lower still (floor 0), reflecting t
hat the two
   problems compound and that complexity dominates.
4. Before penalizing Volume at all, check the hopping/sliding
single-
   element exception above — a moving focus on one element at
a time is
   never a volume violation regardless of how many elements a
re nearby.
""".strip()

OUTPUT_FORMAT_INSTRUCTIONS = """
Respond in EXACTLY this structure, with these literal section
headers:

### RATIONALE
<Write a thorough step-by-step rationale. First identify what
is NEWLY TARGETED in the current frame relative to the previo
us frame according to the Style Adapter, mapping it to specif
```

```
ic XML IDs. Then analyze Volume (how many elements/groups wer
e targeted at this step, and whether that count is digestible
or an overload). Then analyze Complexity (whether any targete
d element is intrinsically complex per the XML signals). Then
explain how you combined them into the final band using the t
ie-break rule.>

### VOLUME_BAND: <integer 0-4, your standalone judgment for V
olume only>
### COMPLEXITY_BAND: <integer 0-4, your standalone judgment f
or Complexity only>
### FINAL_BAND: <integer 0-4, the combined band per the tie-b
reak rule>
### LABEL: <one of: Well-Paced | Minor Overload | Moderate Ov
erload | Major Overload | Severe Overload>

Do not output anything after the LABEL line.
""".strip()

def build_system_instruction():
    return (
        "You are a meticulous evaluator of diagram animation
s, specifically "
        "judging PACING — whether each timestep targets a dig
estible unit of "
        "information or overloads the viewer. You are an expe
rt at reading "
        "hierarchical XML diagram schemas (blocks, standalone
nodes, child nodes, "
        "raster nodes, composite wholes/parts, and edges) and
using that structure "
        "to judge the true visual and cognitive complexity of
what's being "
        "targeted, not just a raw element count. You are stri
ct about overload but "
        "never penalize an animation for being appropriately
```

```
fine-grained or "
        "sparse at a given step. You always ground your reasoning in specific "
        "element ids from the provided XML, never vague visual impressions alone.\n\n"
        + RUBRIC_TEXT
    )

def build_user_prompt(xml_text, timestep_idx, total_steps, prior_summary, style_name="Progressive Reveal"):
    # Fetch the corresponding adapter text, default to Progressive Reveal if not found
    style_adapter_text = STYLE_ADAPTERS.get(style_name, STYLE_ADAPTERS["Progressive Reveal"])

    return (
        f"This is timestep {timestep_idx} of {total_steps} in a '{style_name}' "
        "animation of the diagram below.\n\n"
        "You are given, in order:\n"
        "1. The ORIGINAL fully-revealed diagram image (ground truth layout).\n"
        "2. The PREVIOUS timestep's rendered frame (state before this reveal). "
        "Omitted if this is timestep 1.\n"
        "3. The CURRENT timestep's rendered frame (state after this reveal).\n\n"
        f"--- CUMULATIVE ELEMENTS ALREADY TARGETED BEFORE THIS STEP ---\n"
        f"{prior_summary}\n\n"
        "--- FULL XML DIAGRAM SCHEMA ---\n"
        f"{xml_text}\n\n"
        f"{style_adapter_text}\n\n"
        "Identify what is NEW in the CURRENT frame relative to the PREVIOUS "
        "frame (or, if timestep 1, what the entrypoint consis
```

```
ts of) based strictly "
        "on the Style Adapter definition above, map it onto i
ds in the XML, and score the "
        "VOLUME and COMPLEXITY of that newly-targeted group p
er the rubric.\n\n"
        + OUTPUT_FORMAT_INSTRUCTIONS
    )

# -------------------------------------------------------------
----------------
# Output parsing
# -------------------------------------------------------------
----------------

_BAND_RE = {
    "volume_band": re.compile(r"###\s*VOLUME_BAND:\s*([0-
4])", re.IGNORECASE),
    "complexity_band": re.compile(r"###\s*COMPLEXITY_BAND:\s*
([0-4])", re.IGNORECASE),
    "final_band": re.compile(r"###\s*FINAL_BAND:\s*([0-4])",
re.IGNORECASE),
    "label": re.compile(r"###\s*LABEL:\s*(.+)", re.IGNORECAS
E),
}
_RATIONALE_RE = re.compile(
    r"###\s*RATIONALE\s*(.*?)(?=###\s*VOLUME_BAND)", re.IGNOR
ECASE | re.DOTALL
)

def parse_judge_output(raw_text):
    """
    Parses a judge's raw text output into a structured dict.
Returns
    is_valid=False if FINAL_BAND cannot be recovered, so the
caller can
    log/retry rather than silently poisoning the average.
```

```
    """
    result = {
        "rationale": None,
        "volume_band": None,
        "complexity_band": None,
        "final_band": None,
        "label": None,
        "is_valid": False,
        "raw_output": raw_text,
    }

    rationale_match = _RATIONALE_RE.search(raw_text)
    if rationale_match:
        result["rationale"] = rationale_match.group(1).strip
()

    for key, pattern in _BAND_RE.items():
        m = pattern.search(raw_text)
        if m:
            if key == "label":
                result[key] = m.group(1).strip().splitlines()
[0].strip()
            else:
                result[key] = int(m.group(1))

    if result["final_band"] is not None:
        result["is_valid"] = True
        if not result["label"]:
            result["label"] = BAND_LABELS.get(result["final_b
and"])

    return result

def read_file_safely(filepath):
    if os.path.exists(filepath):
        with open(filepath, "r", encoding="utf-8", errors="re
```

```
place") as f:
            return f.read().strip()
    return ""

# ------------------------------------------------------------
----------------
# Sample layout discovery
# ------------------------------------------------------------
----------------

_IMAGE_EXTS = (".png", ".jpg", ".jpeg")
_FRAMES_DIR_CANDIDATES = ("frames", "animation_frames", "vide
o_frames", "anim_frames")

def locate_sample_files(sample_dir):
    if not os.path.isdir(sample_dir):
        return None, None, None, f"sample_dir does not exist:
{sample_dir}"

    entries = os.listdir(sample_dir)
    top_files = [e for e in entries if os.path.isfile(os.pat
h.join(sample_dir, e))]
    top_dirs = [e for e in entries if os.path.isdir(os.path.j
oin(sample_dir, e))]

    image_files = [f for f in top_files if f.lower().endswith
(_IMAGE_EXTS)]
    original_img = None
    for f in image_files:
        low = f.lower()
        if low.startswith(("original", "gt", "ground_truth",
"source")):
            original_img = os.path.join(sample_dir, f)
            break
    if original_img is None and len(image_files) == 1:
        original_img = os.path.join(sample_dir, image_files
```

```
[0])

    xml_files = [f for f in top_files if f.lower().endswith
(".xml")]
    xml_path = os.path.join(sample_dir, xml_files[0]) if len
(xml_files) >= 1 else None
    if len(xml_files) > 1:
        preferred = [f for f in xml_files if "diagram" in f.l
ower() or "schema" in f.lower()]
        if preferred:
            xml_path = os.path.join(sample_dir, preferred[0])

    frames_dir = None
    for cand in _FRAMES_DIR_CANDIDATES:
        for d in top_dirs:
            if d.lower() == cand:
                frames_dir = os.path.join(sample_dir, d)
                break
        if frames_dir:
            break
    if frames_dir is None and len(top_dirs) == 1:
        frames_dir = os.path.join(sample_dir, top_dirs[0])

    missing = []
    if original_img is None:
        missing.append(f"original image (found top-level imag
es: {image_files or 'none'})")
    if xml_path is None:
        missing.append(f"xml schema file (found top-level .xm
l files: {xml_files or 'none'})")
    if frames_dir is None:
        missing.append(f"frames subdirectory (found subdirect
ories: {top_dirs or 'none'})")

    if missing:
        return original_img, xml_path, frames_dir, "missing:
```

```
" + "; ".join(missing)
    return original_img, xml_path, frames_dir, None
```

Results for Alpha Masking

attachment:e4fc1132-5da2-4638-ab41-2ad8db558d78:CVPR_2025_pipe00004_video_tikz_AM.mp4


```
{
  "sample_dir": "/archive/medha.sen/DAG_test/bench/sample1_AM_TikZ",
  "model": "google/gemma-4-31B-it",
  "n_timesteps": 15,
  "timesteps": [
    {
      "rationale": "The current timestep is the entrypoint reveal. Looking at the frame, the only elements revealed are the labels `views_lbl` (\"Views: $\\xi_{1:K}$\"), `I_1_K_lbl` (\"$I_{1:K}$\"), and `M_1_K_lbl` (\"$M_{1:K}$\"), along with the image stacks `I_stack_1`, `I_stack_2`, `I_stack_3` and `M_stack_1`, `M_stack_2`, `M_stack_3`. \n\n**Volume:** The number of elements is small. While there are multiple raster nodes for the stacks, they visually represent two single conceptual groups (the image stack and the mask stack). This is a very sparse and digestible
```

start to the animation.\n
\n**Complexity:** The elem
ents revealed are simple `
raster_node`s and `text_no
de`s. There are no `block`
structures with multiple c
hildren, no `composite_par
t`s, and no `edge`s reveal
ed in this step. The intri
nsic complexity is very lo
w.\n\n**Combination:** Bot
h Volume and Complexity ar
e band 4. According to the
tie-break rule, the final
band is 4.",
      "volume_band": 4,
      "complexity_band":
4,
      "final_band": 4,
      "label": "Well-Pace
d",
      "is_valid": true,
      "raw_output": "### R
ATIONALE\nThe current time
step is the entrypoint rev
eal. Looking at the frame,
the only elements revealed
are the labels `views_lbl`
(\"Views: $\\xi_{1:K}$\"),
`I_1_K_lbl` (\"$I_{1:K}
$\"), and `M_1_K_lbl`
(\"$M_{1:K}$\"), along wit
h the image stacks `I_stac
k_1`, `I_stack_2`, `I_stac
k_3` and `M_stack_1`, `M_s
tack_2`, `M_stack_3`. \n\n

```
**Volume:** The number of
elements is small. While t
here are multiple raster n
odes for the stacks, they
visually represent two sin
gle conceptual groups (the
image stack and the mask s
tack). This is a very spar
se and digestible start to
the animation.\n\n**Comple
xity:** The elements revea
led are simple `raster_nod
e`s and `text_node`s. Ther
e are no `block` structure
s with multiple children,
no `composite_part`s, and
no `edge`s revealed in thi
s step. The intrinsic comp
lexity is very low.\n\n**C
ombination:** Both Volume
and Complexity are band 4.
According to the tie-break
rule, the final band is
4.\n\n### VOLUME_BAND: 4\n
### COMPLEXITY_BAND: 4\n##
# FINAL_BAND: 4\n### LABE
L: Well-Paced",
      "timestep": 1,
      "frame_path": "/arch
ive/medha.sen/DAG_test/ben
ch/sample1_AM_TikZ/frames/
1_CVPR_2025_pipe00004_fram
es_tikz_AM.png"
    },
    {
      "rationale": "1. **V
```

```
olume Analysis**: In this
timestep, the animation re
veals two elements: the `b
lock` `G_box` (which conta
ins the raster image `G_bo
x_img`) and the `edge` `ar
row_I_G` connecting the pr
eviously revealed input vi
ews to this block. This is
a very small, sparse revea
l. According to the rubri
c, sparsity is never a vio
lation. The volume is well
within digestible limit
s.\n\n2. **Complexity Anal
ysis**: The revealed `bloc
k` `G_box` is relatively s
imple; it contains a singl
e `raster_node` (`G_box_im
g`). While it represents a
3D Gaussian Splatting mode
l, visually it is just one
image in a bounding box. T
he `edge` `arrow_I_G` is a
simple directional relatio
nship. There is no high fa
n-in/fan-out or nested com
plexity that would cause c
ognitive overload.\n\n3. *
*Combining Bands**: Both V
olume and Complexity are r
ated as 4 (Well-Paced). Fo
llowing the tie-break rul
e, the final band is 4.",
      "volume_band": 4,
      "complexity_band":
```

4,
      "final_band": 4,
      "label": "Well-Pace
d",
      "is_valid": true,
      "raw_output": "### R
ATIONALE\n1. **Volume Anal
ysis**: In this timestep,
the animation reveals two
elements: the `block` `G_b
ox` (which contains the ra
ster image `G_box_img`) an
d the `edge` `arrow_I_G` c
onnecting the previously r
evealed input views to thi
s block. This is a very sm
all, sparse reveal. Accord
ing to the rubric, sparsit
y is never a violation. Th
e volume is well within di
gestible limits.\n\n2. **C
omplexity Analysis**: The
revealed `block` `G_box` i
s relatively simple; it co
ntains a single `raster_no
de` (`G_box_img`). While i
t represents a 3D Gaussian
Splatting model, visually
it is just one image in a
bounding box. The `edge` `
arrow_I_G` is a simple dir
ectional relationship. The
re is no high fan-in/fan-o
ut or nested complexity th
at would cause cognitive o
verload.\n\n3. **Combining

```
Bands**: Both Volume and C
omplexity are rated as 4
(Well-Paced). Following th
e tie-break rule, the fina
l band is 4.\n\n### VOLUME
_BAND: 4\n### COMPLEXITY_B
AND: 4\n### FINAL_BAND: 4
\n### LABEL: Well-Paced",
      "timestep": 2,
      "frame_path": "/arch
ive/medha.sen/DAG_test/ben
ch/sample1_AM_TikZ/frames/
2_CVPR_2025_pipe00004_fram
es_tikz_AM.png"
    },
    {
      "rationale": "1. **V
olume Analysis**: The curr
ent timestep reveals three
elements:\n    - An edge:
`arrow_G_D` (the blue arro
w labeled \"Render Depth
\").\n    - A text node: `
D_1_K_lbl` (the label \"D_
{1:K}\").\n    - A group o
f raster nodes representin
g a depth map stack: `D_st
ack_1` (including its chil
d `D_stack_1_img`), `D_sta
ck_2`, and `D_stack_3`.\n
The total number of distin
ct visual units is very lo
w (one arrow, one label, o
ne stack of images). This
is a highly digestible chu
nk. Volume is well-pace
```

```
d.\n\n2. **Complexity Anal
ysis**: \n    - The `D_sta
ck` is a simple group of r
aster nodes. While it cont
ains multiple layers to re
present a \"stack,\" it is
a single conceptual unit
(the rendered depth map
s).\n    - The edge `arrow
_G_D` is a simple directed
relationship from the prev
iously revealed `G_box` to
the new `D_stack_1`.\n
- There are no complex blo
cks with numerous childre
n, no high fan-in/fan-out
edges, and no composite 3D
parts revealed in this ste
p.\n    The complexity is
very low.\n\n3. **Combinin
g**: Both Volume and Compl
exity are band 4. Accordin
g to the tie-break rule, t
he final band is 4.",
      "volume_band": 4,
      "complexity_band":
4,
      "final_band": 4,
      "label": "Well-Pace
d",
      "is_valid": true,
      "raw_output": "### R
ATIONALE\n1. **Volume Anal
ysis**: The current timest
ep reveals three element
s:\n    - An edge: `arrow_
```

```
G_D` (the blue arrow label
ed \"Render Depth\").\n
- A text node: `D_1_K_lbl`
(the label \"D_{1:K}\").\n
- A group of raster nodes
representing a depth map s
tack: `D_stack_1` (includi
ng its child `D_stack_1_im
g`), `D_stack_2`, and `D_s
tack_3`.\n    The total nu
mber of distinct visual un
its is very low (one arro
w, one label, one stack of
images). This is a highly
digestible chunk. Volume i
s well-paced.\n\n2. **Comp
lexity Analysis**: \n    -
The `D_stack` is a simple
group of raster nodes. Whi
le it contains multiple la
yers to represent a \"stac
k,\" it is a single concep
tual unit (the rendered de
pth maps).\n    - The edge
`arrow_G_D` is a simple di
rected relationship from t
he previously revealed `G_
box` to the new `D_stack_1
`.\n    - There are no com
plex blocks with numerous
children, no high fan-in/f
an-out edges, and no compo
site 3D parts revealed in
this step.\n    The comple
xity is very low.\n\n3. **
Combining**: Both Volume a
```

```
nd Complexity are band 4.
According to the tie-break
rule, the final band is
4.\n\n### VOLUME_BAND: 4\n
### COMPLEXITY_BAND: 4\n##
# FINAL_BAND: 4\n### LABE
L: Well-Paced",
      "timestep": 3,
      "frame_path": "/arch
ive/medha.sen/DAG_test/ben
ch/sample1_AM_TikZ/frames/
3_CVPR_2025_pipe00004_fram
es_tikz_AM.png"
    },
    {
      "rationale": "1. **V
olume Analysis**: The curr
ent timestep reveals two d
istinct blocks: `box_I2_M2
` and `box_I1_M1`. Each bl
ock contains a pair of ima
ges (one RGB view and one
binary mask) and their cor
responding labels. In tota
l, 8 small elements (2 lab
els and 2 images per bloc
k) are revealed. This is a
very digestible chunk of i
nformation, as it simply i
ntroduces the \"individual
view images and masks\" me
ntioned in the caption. Vo
lume is well-paced.\n\n2.
**Complexity Analysis**:
\n   - `box_I2_M2` contain
s `I2_lbl`, `I2_img`, `M2_
```

```
lbl`, and `M2_img`.\n   -
`box_I1_M1` contains `I1_l
bl`, `I1_img`, `M1_lbl`, a
nd `M1_img`.\n   While the
se are `block` elements, t
hey are simple pairings of
an image and its mask. The
re are no edges (relations
hips) being revealed in th
is step, and the internal
structure of the blocks is
straightforward. The viewe
r is simply seeing two exa
mples of the same data pai
r. There is no intrinsic c
omplexity here that would
cause overload.\n\n3. **Co
mbination**: Both Volume a
nd Complexity are rated as
4. According to the tie-br
eak rule, the final band i
s 4.",
      "volume_band": 4,
      "complexity_band":
4,
      "final_band": 4,
      "label": "Well-Pace
d",
      "is_valid": true,
      "raw_output": "### R
ATIONALE\n1. **Volume Anal
ysis**: The current timest
ep reveals two distinct bl
ocks: `box_I2_M2` and `box
_I1_M1`. Each block contai
ns a pair of images (one R
```

```
GB view and one binary mas
k) and their corresponding
labels. In total, 8 small
elements (2 labels and 2 i
mages per block) are revea
led. This is a very digest
ible chunk of information,
as it simply introduces th
e \"individual view images
and masks\" mentioned in t
he caption. Volume is well
-paced.\n\n2. **Complexity
Analysis**: \n   - `box_I2
_M2` contains `I2_lbl`, `I
2_img`, `M2_lbl`, and `M2_
img`.\n   - `box_I1_M1` co
ntains `I1_lbl`, `I1_img`,
`M1_lbl`, and `M1_img`.\n
While these are `block` el
ements, they are simple pa
irings of an image and its
mask. There are no edges
(relationships) being reve
aled in this step, and the
internal structure of the
blocks is straightforward.
The viewer is simply seein
g two examples of the same
data pair. There is no int
rinsic complexity here tha
t would cause overload.\n
\n3. **Combination**: Both
Volume and Complexity are
rated as 4. According to t
he tie-break rule, the fin
al band is 4.\n\n### VOLUM
```

```
E_BAND: 4\n### COMPLEXITY_
BAND: 4\n### FINAL_BAND: 4
\n### LABEL: Well-Paced",
      "timestep": 4,
      "frame_path": "/arch
ive/medha.sen/DAG_test/ben
ch/sample1_AM_TikZ/frames/
4_CVPR_2025_pipe00004_fram
es_tikz_AM.png"
    },
    {
      "rationale": "1. **V
olume Analysis**: In this
timestep, the animation re
veals two image elements a
nd two connecting edges. S
pecifically, the images `I
2B_img` and `I1B_img` (whi
ch represent the isolated
background regions) are re
vealed, along with the edg
es `arrow_M2_I2B` and `arr
ow_M1_I1B` that connect th
e masks from the previous
step to these images. This
is a very small, digestibl
e number of elements. Volu
me is well-paced.\n\n2. **
Complexity Analysis**: The
revealed elements are simp
le `raster_node`s (`I2B_im
g`, `I1B_img`) and simple
`edge`s. There are no comp
lex `block` structures, hi
gh fan-in/fan-out nodes, o
r composite parts being in
```

```
troduced here. Each image
is a direct result of the
mask above it, making the
cognitive load minimal. Co
mplexity is well-paced.\n
\n3. **Combination**: Both
Volume and Complexity are
scored as 4. According to
the tie-break rule, the fi
nal band is 4.",
      "volume_band": 4,
      "complexity_band":
4,
      "final_band": 4,
      "label": "Well-Pace
d",
      "is_valid": true,
      "raw_output": "### R
ATIONALE\n1. **Volume Anal
ysis**: In this timestep,
the animation reveals two
image elements and two con
necting edges. Specificall
y, the images `I2B_img` an
d `I1B_img` (which represe
nt the isolated background
regions) are revealed, alo
ng with the edges `arrow_M
2_I2B` and `arrow_M1_I1B`
that connect the masks fro
m the previous step to the
se images. This is a very
small, digestible number o
f elements. Volume is well
-paced.\n\n2. **Complexity
Analysis**: The revealed e
```

```
lements are simple `raster
_node`s (`I2B_img`, `I1B_i
mg`) and simple `edge`s. T
here are no complex `block
` structures, high fan-in/
fan-out nodes, or composit
e parts being introduced h
ere. Each image is a direc
t result of the mask above
it, making the cognitive l
oad minimal. Complexity is
well-paced.\n\n3. **Combin
ation**: Both Volume and C
omplexity are scored as 4.
According to the tie-break
rule, the final band is
4.\n\n### VOLUME_BAND: 4\n
### COMPLEXITY_BAND: 4\n##
# FINAL_BAND: 4\n### LABE
L: Well-Paced",
      "timestep": 5,
      "frame_path": "/arch
ive/medha.sen/DAG_test/ben
ch/sample1_AM_TikZ/frames/
5_CVPR_2025_pipe00004_fram
es_tikz_AM.png"
    },
    {
      "rationale": "1. **V
olume Analysis**: Comparin
g the current frame to the
previous frame, the only n
ew elements revealed are t
he text label `D2_lbl` and
the raster image `D2_img`
(including its child `D2_i
```

```
mg_r`). This is a very sma
ll, focused addition consi
sting of one labeled imag
e. According to the rubri
c, sparsity is never a vio
lation. The volume is well
within digestible limit
s.\n\n2. **Complexity Anal
ysis**: The revealed eleme
nt `D2_img` is a simple ra
ster node. While it has an
incident edge `arrow_I2B_D
2` (from `I2B_img`) and `a
rrow_D2_I12B` (to `I12B_im
g`), these relationships a
re revealed as part of the
logical flow of the diagra
m. The element itself is n
ot a complex `block` with
multiple children, nor doe
s it have an overwhelming
fan-in/fan-out that would
cause cognitive overload.
It is a single, simple dat
a representation (a depth
map).\n\n3. **Combination*
*: Both Volume and Complex
ity are rated as 4. Follow
ing the tie-break rule (Ru
le 1), the final band is
4.",
      "volume_band": 4,
      "complexity_band":
4,
      "final_band": 4,
      "label": "Well-Pace
```

```
d",
      "is_valid": true,
      "raw_output": "### R
ATIONALE\n1. **Volume Anal
ysis**: Comparing the curr
ent frame to the previous
frame, the only new elemen
ts revealed are the text l
abel `D2_lbl` and the rast
er image `D2_img` (includi
ng its child `D2_img_r`).
This is a very small, focu
sed addition consisting of
one labeled image. Accordi
ng to the rubric, sparsity
is never a violation. The
volume is well within dige
stible limits.\n\n2. **Com
plexity Analysis**: The re
vealed element `D2_img` is
a simple raster node. Whil
e it has an incident edge
`arrow_I2B_D2` (from `I2B_
img`) and `arrow_D2_I12B`
(to `I12B_img`), these rel
ationships are revealed as
part of the logical flow o
f the diagram. The element
itself is not a complex `b
lock` with multiple childr
en, nor does it have an ov
erwhelming fan-in/fan-out
that would cause cognitive
overload. It is a single,
simple data representation
(a depth map).\n\n3. **Com
```

```
bination**: Both Volume an
d Complexity are rated as
4. Following the tie-break
rule (Rule 1), the final b
and is 4.\n\n### VOLUME_BA
ND: 4\n### COMPLEXITY_BAN
D: 4\n### FINAL_BAND: 4\n#
## LABEL: Well-Paced",
      "timestep": 6,
      "frame_path": "/arch
ive/medha.sen/DAG_test/ben
ch/sample1_AM_TikZ/frames/
6_CVPR_2025_pipe00004_fram
es_tikz_AM.png"
    },
    {
      "rationale": "1. **V
olume Analysis**: Comparin
g the current frame to the
previous frame, the only n
ew visual element revealed
is the image `I12B_img` (l
abeled $I_{1,2}^B$ in the
diagram). This is a single
raster node. According to
the rubric, revealing a si
ngle element is sparse and
is never a volume violatio
n.\n2. **Complexity Analys
is**: The element `I12B_im
g` is a simple raster nod
e. While it has two incomi
ng edges (`arrow_I1B_I12B`
and `arrow_D2_I12B`) and o
ne outgoing edge (`arrow_I
12B_Mprime1`), these edges
```

```
were already partially pre
sent or are simple directi
onal indicators. The node
itself does not contain ch
ild nodes or complex compo
site parts. It is a digest
ible unit of informatio
n.\n3. **Combining**: Both
Volume and Complexity are
rated as 4 (Well-Paced). F
ollowing the tie-break rul
e, the final band is 4.",
      "volume_band": 4,
      "complexity_band":
4,
      "final_band": 4,
      "label": "Well-Pace
d",
      "is_valid": true,
      "raw_output": "### R
ATIONALE\n1. **Volume Anal
ysis**: Comparing the curr
ent frame to the previous
frame, the only new visual
element revealed is the im
age `I12B_img` (labeled $I
_{1,2}^B$ in the diagram).
This is a single raster no
de. According to the rubri
c, revealing a single elem
ent is sparse and is never
a volume violation.\n2. **
Complexity Analysis**: The
element `I12B_img` is a si
mple raster node. While it
has two incoming edges (`a
```

```
rrow_I1B_I12B` and `arrow_
D2_I12B`) and one outgoing
edge (`arrow_I12B_Mprime1
`), these edges were alrea
dy partially present or ar
e simple directional indic
ators. The node itself doe
s not contain child nodes
or complex composite part
s. It is a digestible unit
of information.\n3. **Comb
ining**: Both Volume and C
omplexity are rated as 4
(Well-Paced). Following th
e tie-break rule, the fina
l band is 4.\n\n### VOLUME
_BAND: 4\n### COMPLEXITY_B
AND: 4\n### FINAL_BAND: 4
\n### LABEL: Well-Paced",
      "timestep": 7,
      "frame_path": "/arch
ive/medha.sen/DAG_test/ben
ch/sample1_AM_TikZ/frames/
7_CVPR_2025_pipe00004_fram
es_tikz_AM.png"
    },
    {
      "rationale": "1. **V
olume Analysis**: Comparin
g the current frame to the
previous frame, only one e
lement is newly revealed:
the raster node `Mprime1_i
mg` (the black and white m
ask image). The text label
`Mprime1_lbl` was already
```

```
present (though faded) in
the previous frame. Reveal
ing a single image is a ve
ry sparse and digestible c
hunk of information. Volum
e is band 4.\n\n2. **Compl
exity Analysis**: The elem
ent `Mprime1_img` is a sim
ple raster node. While it
has an incoming edge from
`I12B_img` and an outgoing
edge to `Mprime1_small_img
` (per the XML schema), th
ese relationships are reve
aled sequentially across t
imesteps. At this specific
moment, the viewer is only
absorbing the result of th
e previous operation (the
mask). There are no comple
x blocks, nested child nod
es, or high-fan-out cluste
rs being dumped simultaneo
usly. Complexity is band
4.\n\n3. **Combination**:
Both Volume and Complexity
are band 4. According to t
he tie-break rule, the fin
al band is 4.",
      "volume_band": 4,
      "complexity_band":
4,
      "final_band": 4,
      "label": "Well-Pace
d",
      "is_valid": true,
```

```
      "raw_output": "### R
ATIONALE\n1. **Volume Anal
ysis**: Comparing the curr
ent frame to the previous
frame, only one element is
newly revealed: the raster
node `Mprime1_img` (the bl
ack and white mask image).
The text label `Mprime1_lb
l` was already present (th
ough faded) in the previou
s frame. Revealing a singl
e image is a very sparse a
nd digestible chunk of inf
ormation. Volume is band
4.\n\n2. **Complexity Anal
ysis**: The element `Mprim
e1_img` is a simple raster
node. While it has an inco
ming edge from `I12B_img`
and an outgoing edge to `M
prime1_small_img` (per the
XML schema), these relatio
nships are revealed sequen
tially across timesteps. A
t this specific moment, th
e viewer is only absorbing
the result of the previous
operation (the mask). Ther
e are no complex blocks, n
ested child nodes, or high
-fan-out clusters being du
mped simultaneously. Compl
exity is band 4.\n\n3. **C
ombination**: Both Volume
and Complexity are band 4.
```

```
According to the tie-break
rule, the final band is
4.\n\n### VOLUME_BAND: 4\n
### COMPLEXITY_BAND: 4\n##
# FINAL_BAND: 4\n### LABE
L: Well-Paced",
      "timestep": 8,
      "frame_path": "/arch
ive/medha.sen/DAG_test/ben
ch/sample1_AM_TikZ/frames/
8_CVPR_2025_pipe00004_fram
es_tikz_AM.png"
    },
    {
      "rationale": "1. **V
olume Analysis**: In this
timestep, the animation re
veals a single visual uni
t: the rendered image and
depth map for the referenc
e view, specifically `Dpri
me1_img` and `Iprime1_img`
(which are visually combin
ed as a single composite o
utput in the diagram). Add
itionally, the text label
`bracket3` (\"Inpainting-g
uided 3DGS Refinement\") i
s highlighted/revealed. Th
e volume is very low (effe
ctively one functional gro
up and one label), which i
s well within the digestib
le range. Sparsity is not
a violation.\n\n2. **Compl
exity Analysis**: The reve
```

```
aled elements are `Dprime1
_img` and `Iprime1_img`. A
ccording to the XML, these
are `raster_node`s. While
they represent a \"render
\" of the current state, t
hey are simple image outpu
ts. There are no complex `
block` structures with mul
tiple children being revea
led, nor are there any new
`edge`s being introduced i
n this specific step that
would create high fan-in/f
an-out cognitive load. The
transition is a simple shi
ft in focus to the start o
f the refinement stage.\n
\n3. **Combination**: Both
Volume and Complexity are
band 4. Following the tie-
break rule, the final band
is 4.",
      "volume_band": 4,
      "complexity_band":
4,
      "final_band": 4,
      "label": "Well-Pace
d",
      "is_valid": true,
      "raw_output": "### R
ATIONALE\n1. **Volume Anal
ysis**: In this timestep,
the animation reveals a si
ngle visual unit: the rend
ered image and depth map f
```

```
or the reference view, spe
cifically `Dprime1_img` an
d `Iprime1_img` (which are
visually combined as a sin
gle composite output in th
e diagram). Additionally,
the text label `bracket3`
(\"Inpainting-guided 3DGS
Refinement\") is highlight
ed/revealed. The volume is
very low (effectively one
functional group and one l
abel), which is well withi
n the digestible range. Sp
arsity is not a violatio
n.\n\n2. **Complexity Anal
ysis**: The revealed eleme
nts are `Dprime1_img` and
`Iprime1_img`. According t
o the XML, these are `rast
er_node`s. While they repr
esent a \"render\" of the
current state, they are si
mple image outputs. There
are no complex `block` str
uctures with multiple chil
dren being revealed, nor a
re there any new `edge`s b
eing introduced in this sp
ecific step that would cre
ate high fan-in/fan-out co
gnitive load. The transiti
on is a simple shift in fo
cus to the start of the re
finement stage.\n\n3. **Co
mbination**: Both Volume a
```

```
nd Complexity are band 4.
Following the tie-break ru
le, the final band is 4.\n
\n### VOLUME_BAND: 4\n###
COMPLEXITY_BAND: 4\n### FI
NAL_BAND: 4\n### LABEL: We
ll-Paced",
      "timestep": 9,
      "frame_path": "/arch
ive/medha.sen/DAG_test/ben
ch/sample1_AM_TikZ/frames/
9_CVPR_2025_pipe00004_fram
es_tikz_AM.png"
    },
    {
      "rationale": "1. **V
olume Analysis**: Comparin
g the current frame to the
previous frame, the new el
ements revealed are:\n
- `Mprime1_small_img` (the
small mask image).\n    -
`arrow_mask_flow` (the das
hed edge connecting `Mprim
e1_img` to `Mprime1_small_
img`).\n    The volume is
very low (two elements), w
hich is well within the di
gestible range. Sparsity i
s not a violation.\n\n2. *
*Complexity Analysis**: \n
- `Mprime1_small_img` is a
simple raster node.\n    -
`arrow_mask_flow` is a sin
gle edge.\n    Neither of
these elements is intrinsi
```

```
cally complex (no blocks w
ith multiple children, no
high fan-in/out, no compos
ite 3D parts). The reveal
is a simple linear progres
sion of a data flow.\n\n3.
**Combining**: Both Volume
and Complexity are band 4.
According to the tie-break
rule, the final band is
4.",
      "volume_band": 4,
      "complexity_band":
4,
      "final_band": 4,
      "label": "Well-Pace
d",
      "is_valid": true,
      "raw_output": "### R
ATIONALE\n1. **Volume Anal
ysis**: Comparing the curr
ent frame to the previous
frame, the new elements re
vealed are:\n    - `Mprime
1_small_img` (the small ma
sk image).\n    - `arrow_m
ask_flow` (the dashed edge
connecting `Mprime1_img` t
o `Mprime1_small_img`).\n
The volume is very low (tw
o elements), which is well
within the digestible rang
e. Sparsity is not a viola
tion.\n\n2. **Complexity A
nalysis**: \n    - `Mprime
1_small_img` is a simple r
```

```
aster node.\n    - `arrow_
mask_flow` is a single edg
e.\n    Neither of these e
lements is intrinsically c
omplex (no blocks with mul
tiple children, no high fa
n-in/out, no composite 3D
parts). The reveal is a si
mple linear progression of
a data flow.\n\n3. **Combi
ning**: Both Volume and Co
mplexity are band 4. Accor
ding to the tie-break rul
e, the final band is 4.\n
\n### VOLUME_BAND: 4\n###
COMPLEXITY_BAND: 4\n### FI
NAL_BAND: 4\n### LABEL: We
ll-Paced",
      "timestep": 10,
      "frame_path": "/arch
ive/medha.sen/DAG_test/ben
ch/sample1_AM_TikZ/frames/
10_CVPR_2025_pipe00004_fra
mes_tikz_AM.png"
    },
    {
      "rationale": "1. **V
olume Analysis**: Comparin
g the current frame to the
previous frame, the newly
revealed elements are:\n
- `DIn1_img` (the rendered
reference image and depth
map pair on the left of th
e inpainting section).\n
- `arrow_mask_to_inpaint`
```

```
(the edge connecting the m
ask `Mprime1_small_img` to
the inpainting process).\n
- The text \"Inpaint\" (as
sociated with the inpainti
ng operation).\n    The vo
lume is very low (one prim
ary image group and one co
nnecting edge), which is w
ell within the digestible
range. Sparsity is not a v
iolation.\n\n2. **Complexi
ty Analysis**: \n    - `DI
n1_img` is a raster node.
While it visually represen
ts a combined image/depth
map, in the XML it is a si
ngle `raster_node`.\n    -
The edge `arrow_mask_to_in
paint` is a simple relatio
nship.\n    - There are no
complex `block` structures
or high fan-in/fan-out nod
es being introduced in thi
s specific step. The viewe
r is simply following the
flow from the mask to the
input image.\n    The comp
lexity is low.\n\n3. **Com
bined Band**: Both Volume
and Complexity are band 4.
According to the tie-break
rule, the final band is
4.",
      "volume_band": 4,
      "complexity_band":
```

```
4,
      "final_band": 4,
      "label": "Well-Pace
d",
      "is_valid": true,
      "raw_output": "### R
ATIONALE\n1. **Volume Anal
ysis**: Comparing the curr
ent frame to the previous
frame, the newly revealed
elements are:\n    - `DIn1
_img` (the rendered refere
nce image and depth map pa
ir on the left of the inpa
inting section).\n    - `a
rrow_mask_to_inpaint` (the
edge connecting the mask `
Mprime1_small_img` to the
inpainting process).\n
- The text \"Inpaint\" (as
sociated with the inpainti
ng operation).\n    The vo
lume is very low (one prim
ary image group and one co
nnecting edge), which is w
ell within the digestible
range. Sparsity is not a v
iolation.\n\n2. **Complexi
ty Analysis**: \n    - `DI
n1_img` is a raster node.
While it visually represen
ts a combined image/depth
map, in the XML it is a si
ngle `raster_node`.\n    -
The edge `arrow_mask_to_in
paint` is a simple relatio
```

```
nship.\n    - There are no
complex `block` structures
or high fan-in/fan-out nod
es being introduced in thi
s specific step. The viewe
r is simply following the
flow from the mask to the
input image.\n    The comp
lexity is low.\n\n3. **Com
bined Band**: Both Volume
and Complexity are band 4.
According to the tie-break
rule, the final band is
4.\n\n### VOLUME_BAND: 4\n
### COMPLEXITY_BAND: 4\n##
# FINAL_BAND: 4\n### LABE
L: Well-Paced",
      "timestep": 11,
      "frame_path": "/arch
ive/medha.sen/DAG_test/ben
ch/sample1_AM_TikZ/frames/
11_CVPR_2025_pipe00004_fra
mes_tikz_AM.png"
    },
    {
      "rationale": "1. **V
olume Analysis**: Comparin
g the current frame to the
previous frame, the newly
revealed elements are:\n
- A raster node: `IP_stack
_1` (the image stack repre
senting projected view
s).\n    - An edge: `arrow
_proj_other` (the blue arr
ow connecting `P1_box` to
```

```
`IP_stack_1`).\n    The vo
lume is very low (two elem
ents), which is well withi
n the digestible range. Sp
arsity is not a violatio
n.\n\n2. **Complexity Anal
ysis**: \n    - `IP_stack_
1` is a raster node that r
epresents a stack of image
s. While it is a \"stac
k,\" it is treated as a si
ngle visual unit in this c
ontext.\n    - `arrow_proj
_other` is a simple direct
ional edge.\n    - Neither
element is a complex `bloc
k` with multiple children,
nor does either have high
fan-in/fan-out that would
create cognitive overload.
The reveal is a simple \"A
leads to B\" transition.\n
\n3. **Combination**: Both
Volume and Complexity are
band 4. According to the t
ie-break rule, the final b
and is 4.",
      "volume_band": 4,
      "complexity_band":
4,
      "final_band": 4,
      "label": "Well-Pace
d",
      "is_valid": true,
      "raw_output": "### R
ATIONALE\n1. **Volume Anal
```

```
ysis**: Comparing the curr
ent frame to the previous
frame, the newly revealed
elements are:\n    - A ras
ter node: `IP_stack_1` (th
e image stack representing
projected views).\n    - A
n edge: `arrow_proj_other`
(the blue arrow connecting
`P1_box` to `IP_stack_1
`).\n    The volume is ver
y low (two elements), whic
h is well within the diges
tible range. Sparsity is n
ot a violation.\n\n2. **Co
mplexity Analysis**: \n
- `IP_stack_1` is a raster
node that represents a sta
ck of images. While it is
a \"stack,\" it is treated
as a single visual unit in
this context.\n    - `arro
w_proj_other` is a simple
directional edge.\n    - N
either element is a comple
x `block` with multiple ch
ildren, nor does either ha
ve high fan-in/fan-out tha
t would create cognitive o
verload. The reveal is a s
imple \"A leads to B\" tra
nsition.\n\n3. **Combinati
on**: Both Volume and Comp
lexity are band 4. Accordi
ng to the tie-break rule,
the final band is 4.\n\n##
```

```
# VOLUME_BAND: 4\n### COMP
LEXITY_BAND: 4\n### FINAL_
BAND: 4\n### LABEL: Well-P
aced",
      "timestep": 12,
      "frame_path": "/arch
ive/medha.sen/DAG_test/ben
ch/sample1_AM_TikZ/frames/
13_CVPR_2025_pipe00004_fra
mes_tikz_AM.png"
    },
    {
      "rationale": "1. **V
olume Analysis**: The curr
ent timestep reveals two p
rimary elements: the `Gpri
me_box` (a block containin
g a raster image and assoc
iated 3D composite parts)
and the `arrow_init` (the
edge connecting `P1_box` t
o `Gprime_box`). This is a
very small, focused additi
on to the diagram, represe
nting a single logical ste
p in the process. The volu
me is well within digestib
le limits.\n\n2. **Complex
ity Analysis**: The reveal
ed element `Gprime_box` is
a `block` containing a `ra
ster_node` (`Gprime_box_im
g`) and several `composite
_part` elements (`Gprime_b
ox_3d1` through `Gprime_bo
x_3d4`). While a block is
```

```
more complex than a standa
lone node, this specific b
lock is a single visual en
tity (a bounded image of a
3D scene). The edge `arrow
_init` is a simple directi
onal link. There is no hig
h fan-in/fan-out or combin
ation of multiple complex
structures that would caus
e cognitive overload. The
reveal is a straightforwar
d transition from the poin
t cloud (`P1_box`) to the
initialized Gaussians (`Gp
rime_box`).\n\n3. **Combin
ation**: Both Volume and C
omplexity are rated as 4.
According to the tie-break
rule, if both sub-criteria
are band 4, the final band
is 4.",
      "volume_band": 4,
      "complexity_band":
4,
      "final_band": 4,
      "label": "Well-Pace
d",
      "is_valid": true,
      "raw_output": "### R
ATIONALE\n1. **Volume Anal
ysis**: The current timest
ep reveals two primary ele
ments: the `Gprime_box` (a
block containing a raster
image and associated 3D co
```

```
mposite parts) and the `ar
row_init` (the edge connec
ting `P1_box` to `Gprime_b
ox`). This is a very smal
l, focused addition to the
diagram, representing a si
ngle logical step in the p
rocess. The volume is well
within digestible limit
s.\n\n2. **Complexity Anal
ysis**: The revealed eleme
nt `Gprime_box` is a `bloc
k` containing a `raster_no
de` (`Gprime_box_img`) and
several `composite_part` e
lements (`Gprime_box_3d1`
through `Gprime_box_3d4`).
While a block is more comp
lex than a standalone nod
e, this specific block is
a single visual entity (a
bounded image of a 3D scen
e). The edge `arrow_init`
is a simple directional li
nk. There is no high fan-i
n/fan-out or combination o
f multiple complex structu
res that would cause cogni
tive overload. The reveal
is a straightforward trans
ition from the point cloud
(`P1_box`) to the initiali
zed Gaussians (`Gprime_box
`).\n\n3. **Combination**:
Both Volume and Complexity
are rated as 4. According
```

```
to the tie-break rule, if
both sub-criteria are band
4, the final band is 4.\n
\n### VOLUME_BAND: 4\n###
COMPLEXITY_BAND: 4\n### FI
NAL_BAND: 4\n### LABEL: We
ll-Paced",
      "timestep": 13,
      "frame_path": "/arch
ive/medha.sen/DAG_test/ben
ch/sample1_AM_TikZ/frames/
14_CVPR_2025_pipe00004_fra
mes_tikz_AM.png"
    },
    {
      "rationale": "1. **V
olume Analysis**: In this
timestep, the animation re
veals a small set of eleme
nts: the `Iprime_stack_1`
(the rendered novel view
s), the `arrow_render_othe
r` edge connecting the Gau
ssian model to these view
s, and the associated text
labels `Render Other views
` and `\u03be2:K`. The vol
ume is very low (a few rel
ated elements), which is w
ell within the digestible
range. Sparsity is not a v
iolation.\n\n2. **Complexi
ty Analysis**: \n    - `Ip
rime_stack_1` is a `raster
_node` representing a stac
k of images. While it repr
```

esents multiple views, it is presented as a single visual unit.\n   - `arrow_render_other` is a simple directed edge.\n   - The text labels are simple.\n   - There are no complex `block` structures or high fan-in/fan-out nodes being introduced here. The elements are simple and logically grouped as a single functional output of the process.\n\n3. **Combination**: Both Volume and Complexity are band 4. According to the tie-break rule, if both are band 4, the final band is 4.",
      "volume_band": 4,
      "complexity_band": 4,
      "final_band": 4,
      "label": "Well-Paced",
      "is_valid": true,
      "raw_output": "### RATIONALE\n1. **Volume Analysis**: In this timestep, the animation reveals a small set of elements: the `Iprime_stack_1` (the rendered novel views), the `arrow_render_other` edge connecting the Gaussian model to these views, and the as

```
sociated text labels `Rend
er Other views` and `\u03b
e2:K`. The volume is very
low (a few related element
s), which is well within t
he digestible range. Spars
ity is not a violation.\n
\n2. **Complexity Analysis
**: \n    - `Iprime_stack_
1` is a `raster_node` repr
esenting a stack of image
s. While it represents mul
tiple views, it is present
ed as a single visual uni
t.\n    - `arrow_render_ot
her` is a simple directed
edge.\n    - The text labe
ls are simple.\n    - Ther
e are no complex `block` s
tructures or high fan-in/f
an-out nodes being introdu
ced here. The elements are
simple and logically group
ed as a single functional
output of the process.\n\n
3. **Combination**: Both V
olume and Complexity are b
and 4. According to the ti
e-break rule, if both are
band 4, the final band is
4.\n\n### VOLUME_BAND: 4\n
### COMPLEXITY_BAND: 4\n##
# FINAL_BAND: 4\n### LABE
L: Well-Paced",
      "timestep": 14,
      "frame_path": "/arch
```

```
ive/medha.sen/DAG_test/ben
ch/sample1_AM_TikZ/frames/
15_CVPR_2025_pipe00004_fra
mes_tikz_AM.png"
    },
    {
      "rationale": "1. **V
olume Analysis**: In this
timestep, the animation re
veals two specific element
s: the red double-headed a
rrow representing the rend
ering loss (`loss_render`)
and the blue arrow represe
nting the reference view r
endering (`arrow_render_re
f`). The volume is very lo
w (two edges), which is we
ll within the digestible r
ange. Sparsity is not a vi
olation.\n\n2. **Complexit
y Analysis**: \n    - `los
s_render` is a simple edge
connecting `DIn1_img` and
`Dprime1_img`.\n    - `arr
ow_render_ref` is a simple
edge connecting `Gprime_bo
x` to `Iprime1_img`.\n
Neither of these elements
is intrinsically complex
(they are not blocks with
multiple children, nor do
they have high fan-in/fan-
out that would overwhelm t
he viewer). They are simpl
e relational indicators.\n
```

```
\n3. **Combining**: Both V
olume and Complexity are r
ated as 4 (Well-Paced). Ac
cording to the tie-break r
ule, the final band is
4.",
      "volume_band": 4,
      "complexity_band":
4,
      "final_band": 4,
      "label": "Well-Pace
d",
      "is_valid": true,
      "raw_output": "### R
ATIONALE\n1. **Volume Anal
ysis**: In this timestep,
the animation reveals two
specific elements: the red
double-headed arrow repres
enting the rendering loss
(`loss_render`) and the bl
ue arrow representing the
reference view rendering
(`arrow_render_ref`). The
volume is very low (two ed
ges), which is well within
the digestible range. Spar
sity is not a violation.\n
\n2. **Complexity Analysis
**: \n    - `loss_render`
is a simple edge connectin
g `DIn1_img` and `Dprime1_
img`.\n    - `arrow_render
_ref` is a simple edge con
necting `Gprime_box` to `I
prime1_img`.\n    Neither
```

```
of these elements is intri
nsically complex (they are
not blocks with multiple c
hildren, nor do they have
high fan-in/fan-out that w
ould overwhelm the viewe
r). They are simple relati
onal indicators.\n\n3. **C
ombining**: Both Volume an
d Complexity are rated as
4 (Well-Paced). According
to the tie-break rule, the
final band is 4.\n\n### VO
LUME_BAND: 4\n### COMPLEXI
TY_BAND: 4\n### FINAL_BAN
D: 4\n### LABEL: Well-Pace
d",
      "timestep": 15,
      "frame_path": "/arch
ive/medha.sen/DAG_test/ben
ch/sample1_AM_TikZ/frames/
16_CVPR_2025_pipe00004_fra
mes_tikz_AM.png"
    }
  ]
}
```

Results for Progressive Reveal

```
{
  "sample_dir": "/archive/
medha.sen/DAG_test/bench/s
ample1_PR_TikZ",
  "model": "google/gemma-4
```

attachment:5177c180-7335-473c-aa5a-8894e11ecd82:CVPR_2025_pipe00010_video_tikz_PR.mp4


```
-31B-it",
  "n_timesteps": 16,
  "timesteps": [
    {
      "rationale": "The cu
rrent timestep is the entr
ypoint reveal. The element
s revealed are:\n- `views_
lbl` (text)\n- `I_1_K_lbl`
(text)\n- `I_stack_1` and
its child `I_stack_1_img`
(raster node representing
the image stack)\n- `M_1_K
_lbl` (text)\n- `M_stack_1
` and its child `M_stack_1
_img` (raster node represe
nting the mask stack)\n\n*
*Volume:** The number of e
lements is very low. It re
veals two primary visual g
roups (the image stack and
the mask stack) along with
their corresponding label
s. This is a sparse, diges
tible start to the animati
on.\n\n**Complexity:** The
elements revealed are simp
le. While `I_stack_1` and
`M_stack_1` contain child
raster nodes, these are ju
st the images themselves,
not complex nested blocks
or high-fan-out nodes. The
re are no edges revealed i
n this step. The cognitive
load is minimal as it simp
```

ly introduces the input da
ta.\n\n**Combination:** Bo
th Volume and Complexity a
re band 4. According to th
e tie-break rule, the fina
l band is 4.",
      "volume_band": 4,
      "complexity_band":
4,
      "final_band": 4,
      "label": "Well-Pace
d",
      "is_valid": true,
      "raw_output": "### R
ATIONALE\nThe current time
step is the entrypoint rev
eal. The elements revealed
are:\n- `views_lbl` (text)
\n- `I_1_K_lbl` (text)\n-
`I_stack_1` and its child
`I_stack_1_img` (raster no
de representing the image
stack)\n- `M_1_K_lbl` (tex
t)\n- `M_stack_1` and its
child `M_stack_1_img` (ras
ter node representing the
mask stack)\n\n**Volume:**
The number of elements is
very low. It reveals two p
rimary visual groups (the
image stack and the mask s
tack) along with their cor
responding labels. This is
a sparse, digestible start
to the animation.\n\n**Com
plexity:** The elements re

```
vealed are simple. While `
I_stack_1` and `M_stack_1`
contain child raster node
s, these are just the imag
es themselves, not complex
nested blocks or high-fan-
out nodes. There are no ed
ges revealed in this step.
The cognitive load is mini
mal as it simply introduce
s the input data.\n\n**Com
bination:** Both Volume an
d Complexity are band 4. A
ccording to the tie-break
rule, the final band is
4.\n\n### VOLUME_BAND: 4\n
### COMPLEXITY_BAND: 4\n##
# FINAL_BAND: 4\n### LABE
L: Well-Paced",
      "timestep": 1,
      "frame_path": "/arch
ive/medha.sen/DAG_test/ben
ch/sample1_PR_TikZ/frames/
1_CVPR_2025_pipe00004_fram
es_tikz_PR_animated.png"
    },
    {
      "rationale": "1. **V
olume Analysis**: In this
timestep, the animation re
veals two new elements: th
e `G_box` (a block contain
ing `G_box_img`) and the `
arrow_I_G` (an edge connec
ting `I_stack_1` to `G_box
`). This is a very small,
```

```
sparse addition to the dia
gram. According to the rub
ric, sparsity is never a v
iolation. The volume is we
ll within digestible limit
s.\n\n2. **Complexity Anal
ysis**: The revealed `G_bo
x` is a `block` containing
a single `raster_node` (`G
_box_img`). While blocks c
an be complex if they cont
ain many child nodes, this
specific block is simple.
The `arrow_I_G` is a singl
e edge representing a stra
ightforward relationship.
There is no high fan-in/fa
n-out or combination of co
mplex structures here. The
cognitive load is very lo
w.\n\n3. **Combining**: Bo
th Volume and Complexity a
re rated at band 4. Follow
ing the tie-break rule (Ru
le 1), the final combined
band is 4.",
      "volume_band": 4,
      "complexity_band":
4,
      "final_band": 4,
      "label": "Well-Pace
d",
      "is_valid": true,
      "raw_output": "### R
ATIONALE\n1. **Volume Anal
ysis**: In this timestep,
```

```
the animation reveals two
new elements: the `G_box`
(a block containing `G_box
_img`) and the `arrow_I_G`
(an edge connecting `I_sta
ck_1` to `G_box`). This is
a very small, sparse addit
ion to the diagram. Accord
ing to the rubric, sparsit
y is never a violation. Th
e volume is well within di
gestible limits.\n\n2. **C
omplexity Analysis**: The
revealed `G_box` is a `blo
ck` containing a single `r
aster_node` (`G_box_img`).
While blocks can be comple
x if they contain many chi
ld nodes, this specific bl
ock is simple. The `arrow_
I_G` is a single edge repr
esenting a straightforward
relationship. There is no
high fan-in/fan-out or com
bination of complex struct
ures here. The cognitive l
oad is very low.\n\n3. **C
ombining**: Both Volume an
d Complexity are rated at
band 4. Following the tie-
break rule (Rule 1), the f
inal combined band is 4.\n
\n### VOLUME_BAND: 4\n###
COMPLEXITY_BAND: 4\n### FI
NAL_BAND: 4\n### LABEL: We
ll-Paced",
```

```
      "timestep": 2,
      "frame_path": "/arch
ive/medha.sen/DAG_test/ben
ch/sample1_PR_TikZ/frames/
2_CVPR_2025_pipe00004_fram
es_tikz_PR_animated.png"
    },
    {
      "rationale": "1. **V
olume Analysis**: The curr
ent timestep reveals three
new elements:\n    - An ed
ge: `arrow_G_D` (the \"Ren
der Depth\" arrow).\n    -
A text label: `D_1_K_lbl
`.\n    - A raster node gr
oup: `D_stack_1` (includin
g its child `D_stack_1_img
`) and the associated stac
k visuals `D_stack_2` and
`D_stack_3`.\n    This is
a very small, focused set
of elements that represent
s a single logical step in
the process (rendering dep
th from the Gaussian mode
l). The volume is well wit
hin digestible limits.\n\n
2. **Complexity Analysis*
*: \n    - The revealed el
ements are simple. The `D_
stack` is a standard raste
r node representation of a
stack of images. \n    - T
he edge `arrow_G_D` has a
simple source (`G_box`) an
```

```
d target (`D_stack_1`), cr
eating a clear, linear rel
ationship.\n    - There ar
e no complex blocks with m
ultiple children or high f
an-in/fan-out edges being
introduced here.\n    The
intrinsic complexity of th
is step is very low.\n\n3.
**Combination**: Both Volu
me and Complexity are rate
d as 4. According to the t
ie-break rule, the final b
and is 4.",
      "volume_band": 4,
      "complexity_band":
4,
      "final_band": 4,
      "label": "Well-Pace
d",
      "is_valid": true,
      "raw_output": "### R
ATIONALE\n1. **Volume Anal
ysis**: The current timest
ep reveals three new eleme
nts:\n    - An edge: `arro
w_G_D` (the \"Render Depth
\" arrow).\n    - A text l
abel: `D_1_K_lbl`.\n    -
A raster node group: `D_st
ack_1` (including its chil
d `D_stack_1_img`) and the
associated stack visuals `
D_stack_2` and `D_stack_3
`.\n    This is a very sma
ll, focused set of element
```

```
s that represents a single
logical step in the proces
s (rendering depth from th
e Gaussian model). The vol
ume is well within digesti
ble limits.\n\n2. **Comple
xity Analysis**: \n    - T
he revealed elements are s
imple. The `D_stack` is a
standard raster node repre
sentation of a stack of im
ages. \n    - The edge `ar
row_G_D` has a simple sour
ce (`G_box`) and target (`
D_stack_1`), creating a cl
ear, linear relationshi
p.\n    - There are no com
plex blocks with multiple
children or high fan-in/fa
n-out edges being introduc
ed here.\n    The intrinsi
c complexity of this step
is very low.\n\n3. **Combi
nation**: Both Volume and
Complexity are rated as 4.
According to the tie-break
rule, the final band is
4.\n\n### VOLUME_BAND: 4\n
### COMPLEXITY_BAND: 4\n##
# FINAL_BAND: 4\n### LABE
L: Well-Paced",
      "timestep": 3,
      "frame_path": "/arch
ive/medha.sen/DAG_test/ben
ch/sample1_PR_TikZ/frames/
3_CVPR_2025_pipe00004_fram
```

```
es_tikz_PR_animated.png"
    },
    {
      "rationale": "1. **V
olume Analysis**: The curr
ent timestep reveals two d
istinct blocks: `box_I2_M2
` and `box_I1_M1`. Each bl
ock contains a pair of ima
ges (one RGB view and one 
mask) and their correspond
ing labels. In total, this
is 8 small elements (2 lab
els and 2 images per bloc
k). This is a very digesti
ble chunk of information, 
as it introduces the \"ind
ividual reference views\" 
mentioned in the caption. 
Volume is well-paced.\n\n
2. **Complexity Analysis*
*: Each revealed element i
s a `block` containing `ch
ild_node`s. Specifically, 
`box_I2_M2` contains `I2_l
bl`, `I2_img`, `M2_lbl`, a
nd `M2_img`. Similarly, `b
ox_I1_M1` contains `I1_lbl
`, `I1_img`, `M1_lbl`, and
`M1_img`. While these are 
blocks, they are simple pa
irings of an image and its
mask. There are no complex
edges (fan-in/out) reveale
d in this step, and the in
ternal structure of the bl
```

```
ocks is straightforward. T
he complexity is low.\n\n
3. **Combining**: Both Vol
ume and Complexity are ban
d 4. According to the tie-
break rule, the final band
is 4.",
      "volume_band": 4,
      "complexity_band":
4,
      "final_band": 4,
      "label": "Well-Pace
d",
      "is_valid": true,
      "raw_output": "### R
ATIONALE\n1. **Volume Anal
ysis**: The current timest
ep reveals two distinct bl
ocks: `box_I2_M2` and `box
_I1_M1`. Each block contai
ns a pair of images (one R
GB view and one mask) and
their corresponding label
s. In total, this is 8 sma
ll elements (2 labels and
2 images per block). This
is a very digestible chunk
of information, as it intr
oduces the \"individual re
ference views\" mentioned
in the caption. Volume is
well-paced.\n\n2. **Comple
xity Analysis**: Each reve
aled element is a `block`
containing `child_node`s.
Specifically, `box_I2_M2`
```

```
contains `I2_lbl`, `I2_img
`, `M2_lbl`, and `M2_img`.
Similarly, `box_I1_M1` con
tains `I1_lbl`, `I1_img`,
`M1_lbl`, and `M1_img`. Wh
ile these are blocks, they
are simple pairings of an
image and its mask. There
are no complex edges (fan-
in/out) revealed in this s
tep, and the internal stru
cture of the blocks is str
aightforward. The complexi
ty is low.\n\n3. **Combini
ng**: Both Volume and Comp
lexity are band 4. Accordi
ng to the tie-break rule,
the final band is 4.\n\n##
# VOLUME_BAND: 4\n### COMP
LEXITY_BAND: 4\n### FINAL_
BAND: 4\n### LABEL: Well-P
aced",
      "timestep": 4,
      "frame_path": "/arch
ive/medha.sen/DAG_test/ben
ch/sample1_PR_TikZ/frames/
4_CVPR_2025_pipe00004_fram
es_tikz_PR_animated.png"
    },
    {
      "rationale": "1. **V
olume Analysis**: The curr
ent timestep reveals four
new elements: two text lab
els (`I2B_lbl`, `I1B_lbl
`), two raster images (`I2
```

```
B_img`, `I1B_img`), and tw
o connecting edges (`arrow
_M2_I2B`, `arrow_M1_I1B`).
This is a very small, dige
stible chunk of informatio
n. The reveal is symmetric
and logically grouped (sho
wing the result of the mas
k application for two diff
erent views). Volume is we
ll-paced.\n\n2. **Complexi
ty Analysis**: The reveale
d elements are simple. The
raster nodes (`I2B_img`, `
I1B_img`) are basic images
without complex internal c
hild structures that would
cause cognitive load. The
edges are simple one-to-on
e mappings from existing n
odes (`M2_img`, `M1_img`)
to the new nodes. There is
no high fan-in/fan-out or
nested block complexity he
re. Complexity is well-pac
ed.\n\n3. **Combination**:
Both Volume and Complexity
are scored as 4. According
to the tie-break rule, the
final band is 4.",
      "volume_band": 4,
      "complexity_band":
4,
      "final_band": 4,
      "label": "Well-Pace
d",
```

```
      "is_valid": true,
      "raw_output": "### R
ATIONALE\n1. **Volume Anal
ysis**: The current timest
ep reveals four new elemen
ts: two text labels (`I2B_
lbl`, `I1B_lbl`), two rast
er images (`I2B_img`, `I1B
_img`), and two connecting
edges (`arrow_M2_I2B`, `ar
row_M1_I1B`). This is a ve
ry small, digestible chunk
of information. The reveal
is symmetric and logically
grouped (showing the resul
t of the mask application
for two different views).
Volume is well-paced.\n\n
2. **Complexity Analysis*
*: The revealed elements a
re simple. The raster node
s (`I2B_img`, `I1B_img`) a
re basic images without co
mplex internal child struc
tures that would cause cog
nitive load. The edges are
simple one-to-one mappings
from existing nodes (`M2_i
mg`, `M1_img`) to the new
nodes. There is no high fa
n-in/fan-out or nested blo
ck complexity here. Comple
xity is well-paced.\n\n3.
**Combination**: Both Volu
me and Complexity are scor
ed as 4. According to the
```

```
tie-break rule, the final
band is 4.\n\n### VOLUME_B
AND: 4\n### COMPLEXITY_BAN
D: 4\n### FINAL_BAND: 4\n#
## LABEL: Well-Paced",
      "timestep": 5,
      "frame_path": "/arch
ive/medha.sen/DAG_test/ben
ch/sample1_PR_TikZ/frames/
5_CVPR_2025_pipe00004_fram
es_tikz_PR_animated.png"
    },
    {
      "rationale": "1. **V
olume Analysis**: The curr
ent timestep reveals four
new elements:\n    - `D2_l
bl` (text node)\n    - `D2
_img` (raster node)\n    -
`I12B_lbl` (text node)\n
- `I12B_img` (raster node)
\n    - `arrow_I2B_D2` (ed
ge)\n    - `arrow_D2_I12B`
(edge)\n    - `arrow_I1B_I
12B` (edge)\n    \n    The
volume is very low. It rev
eals two image-label pairs
and the connecting arrows.
This is a highly digestibl
e chunk of information.\n
\n2. **Complexity Analysis
**: \n    - `D2_img` and `
I12B_img` are simple raste
r nodes.\n    - The edges
(`arrow_I2B_D2`, `arrow_D2
_I12B`, `arrow_I1B_I12B`)
```

```
are straightforward direct
ional relationships.\n
- There are no complex `bl
ock` structures or high-fa
n-out nodes being introduc
ed here. The logic follows
a simple linear flow from
the previously revealed `I
2B_img` and `I1B_img` towa
rd the new `I12B_img`.\n
- Nothing in this step is
intrinsically complex.\n\n
3. **Combination**: Both V
olume and Complexity are w
ell within the limits of a
single digestible step. Fo
llowing the tie-break rul
e, since both are band 4,
the final band is 4.",
      "volume_band": 4,
      "complexity_band":
4,
      "final_band": 4,
      "label": "Well-Pace
d",
      "is_valid": true,
      "raw_output": "### R
ATIONALE\n1. **Volume Anal
ysis**: The current timest
ep reveals four new elemen
ts:\n    - `D2_lbl` (text
node)\n    - `D2_img` (ras
ter node)\n    - `I12B_lbl
` (text node)\n    - `I12B
_img` (raster node)\n    -
`arrow_I2B_D2` (edge)\n
```

```
- `arrow_D2_I12B` (edge)\n
- `arrow_I1B_I12B` (edge)
\n    \n    The volume is
very low. It reveals two i
mage-label pairs and the c
onnecting arrows. This is
a highly digestible chunk
of information.\n\n2. **Co
mplexity Analysis**: \n
- `D2_img` and `I12B_img`
are simple raster nodes.\n
- The edges (`arrow_I2B_D2
`, `arrow_D2_I12B`, `arrow
_I1B_I12B`) are straightfo
rward directional relation
ships.\n    - There are no
complex `block` structures
or high-fan-out nodes bein
g introduced here. The log
ic follows a simple linear
flow from the previously r
evealed `I2B_img` and `I1B
_img` toward the new `I12B
_img`.\n    - Nothing in t
his step is intrinsically
complex.\n\n3. **Combinati
on**: Both Volume and Comp
lexity are well within the
limits of a single digesti
ble step. Following the ti
e-break rule, since both a
re band 4, the final band
is 4.\n\n### VOLUME_BAND:
4\n### COMPLEXITY_BAND: 4
\n### FINAL_BAND: 4\n### L
ABEL: Well-Paced",
```

"timestep": 6,
"frame_path": "/archive/medha.sen/DAG_test/bench/sample1_PR_TikZ/frames/6_CVPR_2025_pipe00004_frames_tikz_PR_animated.png"
},
{
"rationale": "1. **Volume Analysis**: Comparing the current frame to the previous frame, the only new elements revealed are the label `Mprime1_lbl` and the raster node `Mprime1_img` (which contains `Mprime1_img_r`). This is a very small, sparse reveal consisting of one logical unit (a mask image and its label). According to the rubric, sparsity is never a violation.\n2. **Complexity Analysis**: The revealed element `Mprime1_img` is a simple raster node. While it is the target of an edge (`arrow_I12B_Mprime1`), the edge itself is not revealed in this step (it was likely implicit or revealed as part of the node's appearance), and the node does not contain a complex hierarchy of child nodes or composite parts. It is a sin

gle image representing a m
ask. There is no intrinsic
complexity here that would
cause overload.\n3. **Comb
ination**: Both Volume and
Complexity are judged as b
and 4. Following the tie-b
reak rule, the final band
is 4.",
      "volume_band": 4,
      "complexity_band":
4,
      "final_band": 4,
      "label": "Well-Pace
d",
      "is_valid": true,
      "raw_output": "### R
ATIONALE\n1. **Volume Anal
ysis**: Comparing the curr
ent frame to the previous
frame, the only new elemen
ts revealed are the label
`Mprime1_lbl` and the rast
er node `Mprime1_img` (whi
ch contains `Mprime1_img_r
`). This is a very small,
sparse reveal consisting o
f one logical unit (a mask
image and its label). Acco
rding to the rubric, spars
ity is never a violatio
n.\n2. **Complexity Analys
is**: The revealed element
`Mprime1_img` is a simple
raster node. While it is t
he target of an edge (`arr

```
ow_I12B_Mprime1`), the edg
e itself is not revealed i
n this step (it was likely
implicit or revealed as pa
rt of the node's appearanc
e), and the node does not
contain a complex hierarch
y of child nodes or compos
ite parts. It is a single
image representing a mask.
There is no intrinsic comp
lexity here that would cau
se overload.\n3. **Combina
tion**: Both Volume and Co
mplexity are judged as ban
d 4. Following the tie-bre
ak rule, the final band is
4.\n\n### VOLUME_BAND: 4\n
### COMPLEXITY_BAND: 4\n##
# FINAL_BAND: 4\n### LABE
L: Well-Paced",
      "timestep": 7,
      "frame_path": "/arch
ive/medha.sen/DAG_test/ben
ch/sample1_PR_TikZ/frames/
7_CVPR_2025_pipe00004_fram
es_tikz_PR_animated.png"
    },
    {
      "rationale": "1. **V
olume Analysis**: Comparin
g the current frame to the
previous frame, the follow
ing elements are newly rev
ealed:\n    - `Mprime1_sma
ll_img` (the second instan
```

```
ce of the mask in the \"In
painting-guided 3DGS Refin
ement\" section).\n    - `
Mprime1_small_lbl` (the la
bel associated with it).\n
- `arrow_mask_flow` (the d
ashed edge connecting `Mpr
ime1_img` to `Mprime1_smal
l_img`).\n    - `bracket3`
(the text/bracket label
\"Inpainting-guided 3DGS R
efinement\").\n    The vol
ume is very low (three sma
ll elements and one labe
l), which is a highly dige
stible chunk. Sparsity is
not a violation.\n\n2. **C
omplexity Analysis**: \n
- `Mprime1_small_img` is a
simple raster node.\n    -
`arrow_mask_flow` is a sin
gle edge connecting two no
des.\n    - `bracket3` is
a simple text label.\n
None of these elements are
intrinsically complex (no
nested blocks, no high fan
-in/out, no composite 3D p
arts). The transition simp
ly moves the viewer's focu
s from the \"Inferring\" s
ection to the \"Refinement
\" section.\n\n3. **Combin
ing**: Both Volume and Com
plexity are band 4. Accord
ing to the tie-break rule,
```

```
the final band is 4.",
      "volume_band": 4,
      "complexity_band":
4,
      "final_band": 4,
      "label": "Well-Pace
d",
      "is_valid": true,
      "raw_output": "### R
ATIONALE\n1. **Volume Anal
ysis**: Comparing the curr
ent frame to the previous
frame, the following eleme
nts are newly revealed:\n
- `Mprime1_small_img` (the
second instance of the mas
k in the \"Inpainting-guid
ed 3DGS Refinement\" secti
on).\n    - `Mprime1_small
_lbl` (the label associate
d with it).\n    - `arrow_
mask_flow` (the dashed edg
e connecting `Mprime1_img`
to `Mprime1_small_img`).\n
- `bracket3` (the text/bra
cket label \"Inpainting-gu
ided 3DGS Refinement\").\n
The volume is very low (th
ree small elements and one
label), which is a highly
digestible chunk. Sparsity
is not a violation.\n\n2.
**Complexity Analysis**:
\n    - `Mprime1_small_img
` is a simple raster nod
e.\n    - `arrow_mask_flow
```

```
` is a single edge connect
ing two nodes.\n    - `bra
cket3` is a simple text la
bel.\n    None of these el
ements are intrinsically c
omplex (no nested blocks,
no high fan-in/out, no com
posite 3D parts). The tran
sition simply moves the vi
ewer's focus from the \"In
ferring\" section to the
\"Refinement\" section.\n
\n3. **Combining**: Both V
olume and Complexity are b
and 4. According to the ti
e-break rule, the final ba
nd is 4.\n\n### VOLUME_BAN
D: 4\n### COMPLEXITY_BAND:
4\n### FINAL_BAND: 4\n###
LABEL: Well-Paced",
      "timestep": 8,
      "frame_path": "/arch
ive/medha.sen/DAG_test/ben
ch/sample1_PR_TikZ/frames/
8_CVPR_2025_pipe00004_fram
es_tikz_PR_animated.png"
    },
    {
      "rationale": "1. **V
olume Analysis**: Comparin
g the current frame to the
previous frame, the only n
ew elements revealed are t
he text label `Dprime1_lbl
_right` and the correspond
ing raster image `Dprime1_
```

```
img`. This is a very small
number of elements (two re
lated items), which is wel
l within the limits of a d
igestible chunk. Sparsity
is not a violation.\n\n2.
**Complexity Analysis**: T
he revealed element `Dprim
e1_img` is a simple raster
node. It does not contain
child nodes, it is not par
t of a complex block, and
no new edges (relationship
s) are introduced in this
specific timestep. The lab
el `Dprime1_lbl_right` is
a simple text node. There
is no intrinsic complexity
here that would cause view
er overload.\n\n3. **Combi
ning Bands**: Both Volume
and Complexity are scored
as 4 (Well-Paced). Accordi
ng to the tie-break rule,
if both sub-criteria are b
and 4, the final combined
band is 4.",
      "volume_band": 4,
      "complexity_band":
4,
      "final_band": 4,
      "label": "Well-Pace
d",
      "is_valid": true,
      "raw_output": "### R
ATIONALE\n1. **Volume Anal
```

```
ysis**: Comparing the curr
ent frame to the previous 
frame, the only new elemen
ts revealed are the text l
abel `Dprime1_lbl_right` a
nd the corresponding raste
r image `Dprime1_img`. Thi
s is a very small number o
f elements (two related it
ems), which is well within
the limits of a digestible
chunk. Sparsity is not a v
iolation.\n\n2. **Complexi
ty Analysis**: The reveale
d element `Dprime1_img` is
a simple raster node. It d
oes not contain child node
s, it is not part of a com
plex block, and no new edg
es (relationships) are int
roduced in this specific t
imestep. The label `Dprime
1_lbl_right` is a simple t
ext node. There is no intr
insic complexity here that
would cause viewer overloa
d.\n\n3. **Combining Bands
**: Both Volume and Comple
xity are scored as 4 (Well
-Paced). According to the 
tie-break rule, if both su
b-criteria are band 4, the
final combined band is 
4.\n\n### VOLUME_BAND: 4\n
### COMPLEXITY_BAND: 4\n##
# FINAL_BAND: 4\n### LABE
```

```
L: Well-Paced",
      "timestep": 9,
      "frame_path": "/arch
ive/medha.sen/DAG_test/ben
ch/sample1_PR_TikZ/frames/
9_CVPR_2025_pipe00004_fram
es_tikz_PR_animated.png"
    },
    {
      "rationale": "1. **V
olume Analysis**: Comparin
g the current frame to the
previous frame, the follow
ing elements are newly rev
ealed:\n    - `DIn1_lbl`
(text: $D_1^{In}$)\n    -
`DIn1_img` (raster node: t
he rendered depth map)\n
- `IIn1_lbl` (text: $I_1^
{In}$)\n    - `IIn1_img`
(raster node: the rendered
image)\n    - `arrow_inpai
nt` (edge connecting $I'_1
$ to $I_1^{In}$)\n    - `a
rrow_mask_to_inpaint` (edg
e connecting $M'_1$ to $I_
1^{In}$)\n    The volume i
s low to moderate. It reve
als two primary data units
(image and depth map) and
the logical flow (edges) l
eading into them. This is
a digestible chunk.\n\n2.
**Complexity Analysis**:
\n    - The revealed raste
r nodes (`DIn1_img`, `IIn1
```

```
_img`) are simple standalo
ne images.\n    - The edge
s (`arrow_inpaint`, `arrow
_mask_to_inpaint`) represe
nt a clear, single-directi
on logical flow (inpaintin
g process).\n    - There a
re no complex `block` stru
ctures or high-fan-out nod
es being introduced in thi
s step. The complexity is
very low.\n\n3. **Combinin
g**: Both Volume and Compl
exity are well within the
limits of a single digesti
ble step. Following the ti
e-break rule, since both a
re band 4, the final band
is 4.",
      "volume_band": 4,
      "complexity_band":
4,
      "final_band": 4,
      "label": "Well-Pace
d",
      "is_valid": true,
      "raw_output": "### R
ATIONALE\n1. **Volume Anal
ysis**: Comparing the curr
ent frame to the previous
frame, the following eleme
nts are newly revealed:\n
- `DIn1_lbl` (text: $D_1^
{In}$)\n    - `DIn1_img`
(raster node: the rendered
depth map)\n    - `IIn1_lb
```

```
l` (text: $I_1^{In}$)\n
- `IIn1_img` (raster node:
the rendered image)\n    -
`arrow_inpaint` (edge conn
ecting $I'_1$ to $I_1^{In}
$)\n    - `arrow_mask_to_i
npaint` (edge connecting
$M'_1$ to $I_1^{In}$)\n
The volume is low to moder
ate. It reveals two primar
y data units (image and de
pth map) and the logical f
low (edges) leading into t
hem. This is a digestible
chunk.\n\n2. **Complexity
Analysis**: \n    - The re
vealed raster nodes (`DIn1
_img`, `IIn1_img`) are sim
ple standalone images.\n
- The edges (`arrow_inpain
t`, `arrow_mask_to_inpaint
`) represent a clear, sing
le-direction logical flow
(inpainting process).\n
- There are no complex `bl
ock` structures or high-fa
n-out nodes being introduc
ed in this step. The compl
exity is very low.\n\n3. *
*Combining**: Both Volume
and Complexity are well wi
thin the limits of a singl
e digestible step. Followi
ng the tie-break rule, sin
ce both are band 4, the fi
nal band is 4.\n\n### VOLU
```

```
ME_BAND: 4\n### COMPLEXITY
_BAND: 4\n### FINAL_BAND:
4\n### LABEL: Well-Paced",
      "timestep": 10,
      "frame_path": "/arch
ive/medha.sen/DAG_test/ben
ch/sample1_PR_TikZ/frames/
10_CVPR_2025_pipe00004_fra
mes_tikz_PR_animated.png"
    },
    {
      "rationale": "1. **V
olume Analysis**: Comparin
g the current frame to the
previous frame, only one n
ew element is revealed: th
e red double-headed arrow
labeled $\\mathcal{L}_{ren
der}$. In the XML, this co
rresponds to the edge `los
s_render`. Revealing a sin
gle edge is an extremely s
parse and digestible updat
e. Volume is band 4.\n\n2.
**Complexity Analysis**: T
he element `loss_render` i
s a simple edge connecting
two existing raster nodes
(`DIn1_img` and `Dprime1_i
mg`). It does not involve
any complex blocks, child
nodes, or high fan-in/fan-
out structures that would
overload the viewer. The c
ognitive load is minimal a
s it simply introduces a r
```

```
elationship (a loss functi
on) between two already-vi
sible components. Complexi
ty is band 4.\n\n3. **Comb
ination**: Both Volume and
Complexity are band 4. Acc
ording to the tie-break ru
le, the final band is 4.",
      "volume_band": 4,
      "complexity_band":
4,
      "final_band": 4,
      "label": "Well-Pace
d",
      "is_valid": true,
      "raw_output": "### R
ATIONALE\n1. **Volume Anal
ysis**: Comparing the curr
ent frame to the previous
frame, only one new elemen
t is revealed: the red dou
ble-headed arrow labeled
$\\mathcal{L}_{render}$. I
n the XML, this correspond
s to the edge `loss_render
`. Revealing a single edge
is an extremely sparse and
digestible update. Volume
is band 4.\n\n2. **Complex
ity Analysis**: The elemen
t `loss_render` is a simpl
e edge connecting two exis
ting raster nodes (`DIn1_i
mg` and `Dprime1_img`). It
does not involve any compl
ex blocks, child nodes, or
```

```
high fan-in/fan-out struct
ures that would overload t
he viewer. The cognitive l
oad is minimal as it simpl
y introduces a relationshi
p (a loss function) betwee
n two already-visible comp
onents. Complexity is band
4.\n\n3. **Combination**:
Both Volume and Complexity
are band 4. According to t
he tie-break rule, the fin
al band is 4.\n\n### VOLUM
E_BAND: 4\n### COMPLEXITY_
BAND: 4\n### FINAL_BAND: 4
\n### LABEL: Well-Paced",
      "timestep": 11,
      "frame_path": "/arch
ive/medha.sen/DAG_test/ben
ch/sample1_PR_TikZ/frames/
11_CVPR_2025_pipe00004_fra
mes_tikz_PR_animated.png"
    },
    {
      "rationale": "1. **V
olume Analysis**: The curr
ent timestep reveals a sma
ll set of elements: the `P
1_box` (a block containing
`P1_box_img` and associate
d 3D composite parts `P1_b
ox_3d1` through `P1_box_3d
4`), the label `P1_lbl`, a
nd the edge `arrow_depth_p
roj` connecting `DIn1_img`
to the box. This is a very
```

small, focused addition to the diagram. The volume is well within digestible limits.\n\n2. **Complexity Analysis**: The revealed element `P1_box` is a block with several internal components (a raster image and 3D composite parts), which makes it slightly more complex than a standalone node. However, it is revealed as a single logical unit (the 3D point cloud representation). The accompanying edge `arrow_depth_proj` is a simple relationship. There are no other complex elements revealed in this step to compound the cognitive load. The complexity is low and appropriate for a single step.\n\n3. **Combining**: Both Volume and Complexity are judged as band 4. According to the tie-break rule, the final band is 4.",
      "volume_band": 4,
      "complexity_band": 4,
      "final_band": 4,
      "label": "Well-Paced",
      "is_valid": true,
      "raw_output": "### R

```
ATIONALE\n1. **Volume Anal
ysis**: The current timest
ep reveals a small set of
elements: the `P1_box` (a
block containing `P1_box_i
mg` and associated 3D comp
osite parts `P1_box_3d1` t
hrough `P1_box_3d4`), the
label `P1_lbl`, and the ed
ge `arrow_depth_proj` conn
ecting `DIn1_img` to the b
ox. This is a very small,
focused addition to the di
agram. The volume is well
within digestible limit
s.\n\n2. **Complexity Anal
ysis**: The revealed eleme
nt `P1_box` is a block wit
h several internal compone
nts (a raster image and 3D
composite parts), which ma
kes it slightly more compl
ex than a standalone node.
However, it is revealed as
a single logical unit (the
3D point cloud representat
ion). The accompanying edg
e `arrow_depth_proj` is a
simple relationship. There
are no other complex eleme
nts revealed in this step
to compound the cognitive
load. The complexity is lo
w and appropriate for a si
ngle step.\n\n3. **Combini
ng**: Both Volume and Comp
```

```
lexity are judged as band
4. According to the tie-br
eak rule, the final band i
s 4.\n\n### VOLUME_BAND: 4
\n### COMPLEXITY_BAND: 4\n
### FINAL_BAND: 4\n### LAB
EL: Well-Paced",
      "timestep": 12,
      "frame_path": "/arch
ive/medha.sen/DAG_test/ben
ch/sample1_PR_TikZ/frames/
12_CVPR_2025_pipe00004_fra
mes_tikz_PR_animated.png"
    },
    {
      "rationale": "1. **V
olume Analysis**: Comparin
g the current frame to the
previous frame, the follow
ing elements are newly rev
ealed:\n    - `IP_stack_1`
(and its child `IP_stack_1
_img`): The first image in
the \"Project to Other Vie
ws\" stack.\n    - `IP_lbl
`: The text label \"$I^P_
{2:K}$\".\n    - `arrow_pr
oj_other`: The blue edge c
onnecting the 3D point clo
ud `P1_box` to the image s
tack.\n    - The text labe
l \"Project to Other Views
\".\n    The volume is ver
y low (one image, one labe
l, one arrow). This is a s
parse, digestible chunk.\n
```

```
\n2. **Complexity Analysis
**: \n    - `IP_stack_1` i
s a simple raster node.\n
- `arrow_proj_other` is a
single relationship connec
ting a previously revealed
block (`P1_box`) to a new
node.\n    - There are no
complex blocks or high fan
-in/fan-out nodes being in
troduced in this step. The
intrinsic complexity is ve
ry low.\n\n3. **Combinatio
n**: Both Volume and Compl
exity are band 4. Accordin
g to the tie-break rule, t
he final band is 4.",
      "volume_band": 4,
      "complexity_band":
4,
      "final_band": 4,
      "label": "Well-Pace
d",
      "is_valid": true,
      "raw_output": "### R
ATIONALE\n1. **Volume Anal
ysis**: Comparing the curr
ent frame to the previous
frame, the following eleme
nts are newly revealed:\n
- `IP_stack_1` (and its ch
ild `IP_stack_1_img`): The
first image in the \"Proje
ct to Other Views\" stac
k.\n    - `IP_lbl`: The te
xt label \"$I^P_{2:K}
```

```
$\".\n    - `arrow_proj_ot
her`: The blue edge connec
ting the 3D point cloud `P
1_box` to the image stac
k.\n    - The text label 
\"Project to Other Views
\".\n    The volume is ver
y low (one image, one labe
l, one arrow). This is a s
parse, digestible chunk.\n
\n2. **Complexity Analysis
**: \n    - `IP_stack_1` i
s a simple raster node.\n 
- `arrow_proj_other` is a 
single relationship connec
ting a previously revealed
block (`P1_box`) to a new 
node.\n    - There are no 
complex blocks or high fan
-in/fan-out nodes being in
troduced in this step. The
intrinsic complexity is ve
ry low.\n\n3. **Combinatio
n**: Both Volume and Compl
exity are band 4. Accordin
g to the tie-break rule, t
he final band is 4.\n\n###
VOLUME_BAND: 4\n### COMPLE
XITY_BAND: 4\n### FINAL_BA
ND: 4\n### LABEL: Well-Pac
ed",
      "timestep": 13,
      "frame_path": "/arch
ive/medha.sen/DAG_test/ben
ch/sample1_PR_TikZ/frames/
13_CVPR_2025_pipe00004_fra
```

```
mes_tikz_PR_animated.png"
    },
    {
      "rationale": "1. **V
olume Analysis**: In this
timestep, the animation re
veals two primary componen
ts: the `Gprime_box` (whic
h includes its `raster_nod
e` and associated `composi
te_part` 3D elements) and
the `arrow_init` edge conn
ecting `P1_box` to `Gprime
_box`. Additionally, the c
aption text at the bottom
is updated. The volume is
very low\u2014essentially
one logical group (the new
Gaussian box) and one conn
ecting arrow. This is a hi
ghly digestible chunk.\n\n
2. **Complexity Analysis*
*: The `Gprime_box` is a `
block` containing a `raste
r_node` and several `compo
site_part` elements (`Gpri
me_box_3d1` through `Gprim
e_box_3d4`) to create the
3D effect. While a block i
s more complex than a stan
dalone node, this specific
block is a single visual e
ntity (a 3D bounding box w
ith an image inside). The
`arrow_init` is a simple e
dge. There is no high fan-
```

```
in/fan-out or combination
of multiple complex struct
ures here. The complexity
is low and well within the
limits of a single step.\n
\n3. **Combination**: Both
Volume and Complexity are
judged as band 4. Accordin
g to the tie-break rule, i
f both are band 4, the fin
al band is 4.",
      "volume_band": 4,
      "complexity_band":
4,
      "final_band": 4,
      "label": "Well-Pace
d",
      "is_valid": true,
      "raw_output": "### R
ATIONALE\n1. **Volume Anal
ysis**: In this timestep,
the animation reveals two
primary components: the `G
prime_box` (which includes
its `raster_node` and asso
ciated `composite_part` 3D
elements) and the `arrow_i
nit` edge connecting `P1_b
ox` to `Gprime_box`. Addit
ionally, the caption text
at the bottom is updated.
The volume is very low\u20
14essentially one logical
group (the new Gaussian bo
x) and one connecting arro
w. This is a highly digest
```

```
ible chunk.\n\n2. **Comple
xity Analysis**: The `Gpri
me_box` is a `block` conta
ining a `raster_node` and
several `composite_part` e
lements (`Gprime_box_3d1`
through `Gprime_box_3d4`)
to create the 3D effect. W
hile a block is more compl
ex than a standalone node,
this specific block is a s
ingle visual entity (a 3D
bounding box with an image
inside). The `arrow_init`
is a simple edge. There is
no high fan-in/fan-out or
combination of multiple co
mplex structures here. The
complexity is low and well
within the limits of a sin
gle step.\n\n3. **Combinat
ion**: Both Volume and Com
plexity are judged as band
4. According to the tie-br
eak rule, if both are band
4, the final band is 4.\n
\n### VOLUME_BAND: 4\n###
COMPLEXITY_BAND: 4\n### FI
NAL_BAND: 4\n### LABEL: We
ll-Paced",
      "timestep": 14,
      "frame_path": "/arch
ive/medha.sen/DAG_test/ben
ch/sample1_PR_TikZ/frames/
14_CVPR_2025_pipe00004_fra
mes_tikz_PR_animated.png"
```

```
    },
    {
      "rationale": "1. **Volume Analysis**: Comparing the current frame to the previous frame, the following elements are newly revealed:\n    - `Iprime_stack_1`, `Iprime_stack_2`, `Iprime_stack_3` (the stack of images for $I'_{2:K}$)\n    - `Iprime_lbl` (the label $I'_{2:K}$)\n    - `arrow_render_other` (the edge from `Gprime_box` to `Iprime_stack_1`)\n    - `arrow_render_ref` (the edge from `Gprime_box` to `Iprime1_img`)\n    - The text \"Render Other views $\\xi_{2:K}$\" and \"Render Reference view $\\xi_1$\" (associated with the edges).\n    The volume is low; it is a small group of related elements (one image stack, one label, and two outgoing edges from a previously revealed node). This is a very digestible chunk.\n\n2. **Complexity Analysis**: \n    - The image stack (`Iprime_stack_1` etc.) is a simple raster group.\n    - The edges (`arrow_render_other`, `arrow
```

```
_render_ref`) are simple d
irectional relationship
s.\n    - There are no com
plex `block` structures or
high-fan-in/out nodes bein
g introduced that would ca
use cognitive overload. Th
e edges simply complete th
e flow from the `Gprime_bo
x` (the refined 3DGS mode
l) to the final rendered o
utputs.\n    - The complex
ity is low.\n\n3. **Combin
ed Band**: Both Volume and
Complexity are band 4. Acc
ording to the tie-break ru
le, the final band is 4.",
      "volume_band": 4,
      "complexity_band":
4,
      "final_band": 4,
      "label": "Well-Pace
d",
      "is_valid": true,
      "raw_output": "### R
ATIONALE\n1. **Volume Anal
ysis**: Comparing the curr
ent frame to the previous
frame, the following eleme
nts are newly revealed:\n
- `Iprime_stack_1`, `Iprim
e_stack_2`, `Iprime_stack_
3` (the stack of images fo
r $I'_{2:K}$)\n    - `Ipri
me_lbl` (the label $I'_{2:
K}$)\n    - `arrow_render_
```

```
other` (the edge from `Gpr
ime_box` to `Iprime_stack_
1`)\n    - `arrow_render_r
ef` (the edge from `Gprime
_box` to `Iprime1_img`)\n
- The text \"Render Other
views $\\xi_{2:K}$\" and
\"Render Reference view
$\\xi_1$\" (associated wit
h the edges).\n    The vol
ume is low; it is a small
group of related elements
(one image stack, one labe
l, and two outgoing edges
from a previously revealed
node). This is a very dige
stible chunk.\n\n2. **Comp
lexity Analysis**: \n    -
The image stack (`Iprime_s
tack_1` etc.) is a simple
raster group.\n    - The e
dges (`arrow_render_other
`, `arrow_render_ref`) are
simple directional relatio
nships.\n    - There are n
o complex `block` structur
es or high-fan-in/out node
s being introduced that wo
uld cause cognitive overlo
ad. The edges simply compl
ete the flow from the `Gpr
ime_box` (the refined 3DGS
model) to the final render
ed outputs.\n    - The com
plexity is low.\n\n3. **Co
mbined Band**: Both Volume
```

```
and Complexity are band 4.
According to the tie-break
rule, the final band is
4.\n\n### VOLUME_BAND: 4\n
### COMPLEXITY_BAND: 4\n##
# FINAL_BAND: 4\n### LABE
L: Well-Paced",
      "timestep": 15,
      "frame_path": "/arch
ive/medha.sen/DAG_test/ben
ch/sample1_PR_TikZ/frames/
15_CVPR_2025_pipe00004_fra
mes_tikz_PR_animated.png"
    },
    {
      "rationale": "1. **V
olume Analysis**: The only
new element revealed in th
is timestep is a single ed
ge: `loss_cross` (the red
double-headed arrow connec
ting `IP_stack_1` and `Ipr
ime_stack_1`). Revealing a
single element is the defi
nition of sparsity, which
is never a violation. The
volume is perfectly digest
ible.\n2. **Complexity Ana
lysis**: The element `loss
_cross` is a simple edge.
While it connects two exis
ting stacks of images, the
edge itself does not intro
duce a complex sub-structu
re (like a block with many
children) or a high-degree
```

```
node that would overload t
he viewer. It simply estab
lishes a relationship betw
een two previously reveale
d components. The complexi
ty is very low.\n3. **Comb
ination**: Both Volume and
Complexity are rated as 4.
According to the tie-break
rule, if both are band 4,
the final band is 4.",
      "volume_band": 4,
      "complexity_band":
4,
      "final_band": 4,
      "label": "Well-Pace
d",
      "is_valid": true,
      "raw_output": "### R
ATIONALE\n1. **Volume Anal
ysis**: The only new eleme
nt revealed in this timest
ep is a single edge: `loss
_cross` (the red double-he
aded arrow connecting `IP_
stack_1` and `Iprime_stack
_1`). Revealing a single e
lement is the definition o
f sparsity, which is never
a violation. The volume is
perfectly digestible.\n2.
**Complexity Analysis**: T
he element `loss_cross` is
a simple edge. While it co
nnects two existing stacks
of images, the edge itself
```

```
does not introduce a compl
ex sub-structure (like a b
lock with many children) o
r a high-degree node that
would overload the viewer.
It simply establishes a re
lationship between two pre
viously revealed component
s. The complexity is very
low.\n3. **Combination**:
Both Volume and Complexity
are rated as 4. According
to the tie-break rule, if
both are band 4, the final
band is 4.\n\n### VOLUME_B
AND: 4\n### COMPLEXITY_BAN
D: 4\n### FINAL_BAND: 4\n#
## LABEL: Well-Paced",
      "timestep": 16,
      "frame_path": "/arch
ive/medha.sen/DAG_test/ben
ch/sample1_PR_TikZ/frames/
16_CVPR_2025_pipe00004_fra
mes_tikz_PR_animated.png"
    }
  ]
}
```